

Theoretically, it's possible that mean reversion of equity volatilities, rather than the conditional means of equity returns, could produce a downward-sloping cost of insurance over time. But in the real world, it doesn't happen. In Panel C of Figure 5.3, I compute shortfall costs using a model of time-varying volatility developed by Heston (1993). Heston's model specifies stock volatility as mean-reverting, and there is much more predictability in stock volatilities than in conditional means of stock returns. Panel C shows that the cost of insurance with time-varying volatility also increases over time and is quite similar to the cost in Panel B with constant volatility. I plot the cost of insurance using two volatility levels: the long-run steady state volatility of 19% through the 1926–2012 sample and a starting level of volatility of 60%, which we experienced during the Great Depression and the more recent financial crisis. In both cases, stocks are still risky over the long run.

Nevertheless, even though the conventional wisdom that stocks are less risky over the long run is incorrect, the advice that financial advisors give can be appropriate. To state this rigorously requires using . . .

3.3. LIFE-CYCLE MODELS

Individuals earn wages during their working lives, but this flow of income ceases when they retire. Thus, people have to save during their working lives so that they can support themselves during retirement. Economists use *life-cycle models* to study the consumption, savings, and investment decisions of people as they age. The first such model was developed by Franco Modigliani and his student Richard Brumberg. They wrote two papers, one published in 1954 and the other in 1980. Brumberg died early from heart disease, and Modigliani worked on the second paper again only when it was published in a volume of his collected papers.²⁰

Life-cycle models contain at least two periods: an accumulation period in which individuals work and save and a decumulation period during which individuals draw down their savings and so dis-save. There are sometimes other periods, like early versus late retirement, being a young worker and then a middle-aged worker, and so. Individuals tailor their consumption, savings, and investment decisions to their needs at different ages, and pensions enable agents to *smooth consumption* over their life cycles.

Modigliani and Brumberg also created *overlapping generations* (OLG) economies where individuals each follow life-cycle models, and there are different cohorts or generations. In the setting of our simple two-period life-cycle model, there are two generations alive at each time. One generation is young and working, and their parents in the other generation are old and retired. In the next period, a new young generation is born and the previous young generation ages and becomes old. The previous old generation leaves the economy through death. In OLG

²⁰ As recounted by Deaton (2005).